Rai Mai Phao (Zero-Burn Farm) — Team Briefing

*Everything the team should know before talking to visitors and judges. All prices, yields, costs and PM2.5 figures are **prototype estimates** to be tested on a real machine.*

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1. The project in one breath

Farmers burn crop residue because it is the **cheapest option**. We make residue **worth money**: our 3-in-1 machine **KhumKha** ("worth it" in Thai) in a **village station** turns it into **animal feed, fuel briquettes or compost material**. Farmers sell through **LINE**, and a **local driver** collects it from the farm, like Grab. **Farmers earn, drivers get work, buyers get cheap raw material, everyone gets cleaner air.**

2. Numbers to remember

The problem (GISTDA, 2026)

Fact	Number
Agricultural area burned in Thailand, 2026	**26.0 million rai** (record high)
North	**16.17 M rai** (most burning, highest lung-cancer rate per person)
Isan (Northeast)	5.59 M rai
Central	1.93 M rai (corn burning up 760%)
West	1.65 M rai (rice burning up 40×)
East	0.63 M rai (lung cancer rising fastest, Sa Kaeo)
South	almost no burning (haze comes from Indonesian peat fires)
PM2.5 avoided	**~8 g per kg** of residue not burned (estimate)

The machine: KhumKha

Fact	Number
Name	**KhumKha** (คุ้มค่า) = "worth it"
Size	**85 × 180 × 60 cm** (W × H × D), about a water dispenser
Shredder	hammer mill, ~3 kW
A — Pellet	animal feed / fuel pellets · yield **85%** · ~60 min per 100 kg
B — Briquette	fuel briquettes (replaces charcoal) · yield **80%** · ~75 min per 100 kg + 15 min warm-up
C — Chop	compost / mushroom substrate / mulch · yield **~90%** · ~15 min per 100 kg
Maintenance	every **200 working hours** (blades, dies)
Cost per machine	**~200,000 THB** (range 150,000–280,000)

Prices

Item	Price
Buy price: corn residue, rice straw, orchard prunings	**1.00 THB/kg**
Buy price: sugarcane leaves	0.80 THB/kg
Moisture ≤ 15%	full price
Moisture 15–20%	−10%
Moisture > 20%	**not accepted** (dry it on the station's yard first)
Processing fee if the farmer takes it home	A 1.20 · B 1.50 · C 0.40 THB/kg
Pickup fee	**50 THB + 10 THB/km + 0.10 THB/kg**, radius 10 km
Pickup allowed	**≥ 300 kg** or a full vehicle load (pickup truck or e-taen tractor)
Farmer pays for pickup (when selling to us)	**at most 30% of the sale**; our **pickup support fund** pays the rest
Driver earns	**90%** of the full pickup fee, always
Platform fee	**10%** of the pickup fee

Worked examples

- **Sell 1,000 kg of dry corn residue, 6 km away:** sale 1,000 THB · pickup 50 + 60 + 100 = 210 THB (under 30%) → **farmer nets 790 THB** · driver 189 · platform 21.
- **Sell 300 kg, 6 km away:** sale 300 THB · pickup 140 THB · farmer pays 90 (30%) · fund pays 50 → **farmer nets 210 THB** · driver 126.

Business

Fact	Number
Rough payback (illustration)	400 kg/day × 25 days = 10 t/month × ~1.2 THB/kg margin ≈ 12,000 THB/month → **~17 months**
Village station	**~2 rai**, works like a gas station
Software running cost (prototype)	**0 THB/month**
SDGs	**3, 8, 12, 13** (also 7 and 15)

3. How it works (5 steps)

1. **Book in LINE**: the farmer reports residue type and weight, and chooses to bring it or book a pickup.
2. **Village pickup**: a registered local driver accepts the job and collects it from the farm.
3. **Weigh, photo, moisture check** at the station: the farmer's QR code is scanned at the machine.
4. **Process** in mode A, B or C.
5. **Farmer chooses**: sell to us, consign, or take it home. A receipt arrives in LINE automatically.

Station (8 parts): truck scale · covered drying yard · machine shed · solar roof · lot storage · shop & office · fire safety · parking. Start with **one machine per village**, add more as supply grows.

4. How the machine works (top to bottom)

Height	Part	Job
140–180 cm	hopper with safety lid, hand guard, magnet, moisture sensor, touch screen, QR scanner, camera	receive, check, identify the lot
100–140 cm	hammer mill shredder + 2 screens	shred
85–100 cm	motorised diverter	send to A / B / C
30–85 cm	A flat-die pellet press · C straight chute · B heated screw press + cutter	process
0–30 cm	3 output trays with scales	weigh output, compute yield
Back	dust fan + bag, control box, vents	dust, heat, safety

Known limits: one mode at a time (solved with a queue) · switching from fuel to feed needs a flush run · if the shredder fails the whole machine stops (spare blades + automatic technician alert).

5. Business model (BMC)

Block	Content
Customer segments	farmers selling residue · buyers of 6 products · drivers (service providers)
Value propositions	farmers: residue earns money, pickup at the farm · buyers: consistent, traceable supply · community: clean air
Channels	LINE OA + rich menu · community drying yard · village leaders
Customer relationships	LINE OA · transparent receipts · public dashboard
Revenue streams	**product sales (main)** · processing fees · 10% platform fee on pickups · future: CSR sponsors, local-government clean-air budgets, carbon credits
Key activities	buy, process, sell · run the LINE platform · maintain machines
Key resources	KhumKha 3-in-1 machine · LINE system + data · driver network · brand
Key partners	community enterprises · local government · university · mushroom farms, dairy co-ops, longan dryers · local drivers
Cost structure	machine (~200k THB) · electricity (cut by solar) · residue purchases · maintenance · labour · pickup support fund

Who benefits: farmers (income) · village drivers (side income) · buyers (cheap, steady raw material) · community (less PM2.5, fewer respiratory illnesses) · local government (measurable tool to cut burning) · environment (less carbon, residue back to soil). **Triple bottom line:** People = income · Planet = PM2.5 avoided · Profit = margin + fees.

6. SDGs

SDG	Link to the project
3 Good health and well-being	no burning → less PM2.5 → fewer respiratory illnesses and lung cancers
8 Decent work and economic growth	farmers earn from residue; villagers earn as drivers
12 Responsible consumption and production	residue becomes feed, fuel and compost; nothing is wasted (circular economy)
13 Climate action	less carbon from open burning, measured on the dashboard
Also 7 · 15	biomass briquettes + solar roof · fewer wildfires, residue returns to the soil

7. Scripts

30 seconds (passers-by)

> **"Ever been stuck in the smoke haze in the dry season?"** *(wait for the nod)* This year Thailand burned **26 million rai** of crop residue, a record, because burning is the cheapest option for farmers. So we built **KhumKha**, Thai for "worth it": a machine the size of a water dispenser that turns crop waste into **animal feed, fuel briquettes or compost**, all run from **LINE**. Farmers get paid instead of burning, villagers earn as drivers, and everyone breathes cleaner air. We hit **four SDGs**: 3, 8, 12 and 13. **Want to try? Scan this QR, it takes 10 seconds.**

2 minutes (visitors who stop)

1. **Board 1, problem (20 s):** "The North burned 16 million rai, the most in the country, with the highest lung-cancer rate. Farmers don't burn because they don't know better. They burn because there's no cheaper choice."
2. **Board 2 + iPad, machine (40 s):** "This is KhumKha, our 3-in-1 machine. The name means 'worth it' in Thai. Residue goes in the top and gets shredded, then A makes animal feed, B makes fuel briquettes, C makes compost material. It weighs, photographs and checks moisture, and the receipt pops up in LINE. Fully transparent." *(hand over the iPad: X-ray slider, Show workflow)*
3. **Board 3, station (40 s):** "Each village gets a station that works like a gas station. Farmers book in LINE, a local driver picks up like Grab, it's weighed and processed, and the farmer chooses to sell, consign or take it home. We pay about 1 baht per kg, and farmers pay at most 30% of the sale for pickup."
4. **Close (20 s):** "Farmers earn, drivers get work, buyers get cheap raw material, everyone gets cleaner air. That's SDG 3, 8, 12 and 13." *(hand over the phone)* "Try selling some residue in LINE."

4 minutes (judges)

Part	Show	Say	Time
Open	boards	"We're Rai Mai Phao, the Zero-Burn Farm. We make crop waste worth more than burning it."	10 s
Problem	board 1	"26 million rai burned in 2026, a record. The North burned 16 million and has the highest lung-cancer rate. Burning is simply the cheapest option."	30 s
Solution + demo	board 2 + **iPad in the judge's hands**	"This is KhumKha, Thai for 'worth it'. Our 3-in-1 machine shreds residue, then makes feed, briquettes or compost. It weighs, photographs and checks moisture, then sends a receipt to LINE." *(Show workflow, X-ray)*	50 s
System	board 3	"One station per village, like a gas station. Farmers book in LINE and a local driver collects, like Grab. Farmers pay at most 30% of the sale for pickup, and our fund covers the rest."	30 s
Business	iPad → **Business Model**	"Three revenue streams: product sales, processing fees, and 10% of pickup fees. A machine costs about 200,000 baht, with a rough payback of 17 months."	40 s
Sustainability	SDG strip on board 2	"SDG 3 cleaner air, 8 income, 12 nothing wasted, 13 less carbon. People, Planet and Profit together."	20 s
Close	phone with LINE	"Want to try? Sell some residue in LINE, and the receipt arrives in seconds."	20 s

Always add: *"These are prototype estimates; our next step is testing a real machine."*

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8. Question bank

Problem & idea

- **Why do farmers burn?** It's the cheapest and fastest way to clear the field before the next crop. Ploughing it in costs money and earns nothing.
- **Why not just ban burning?** Bans exist, but without a cheaper alternative farmers still burn. We give them a reason not to: income.
- **Who is your customer?** Two sides: farmers who sell residue, and buyers of the products (mushroom farms, dairy co-ops, animal farms, factories with boilers).

Machine

- **What's the machine called?** KhumKha, Thai for "worth it": it makes not burning worth more than burning. The brand is Rai Mai Phao / Zero-Burn Farm.
- **What does the machine make?** A: animal feed or fuel pellets, B: fuel briquettes, C: chopped material for compost, mushrooms or mulch.
- **How fast is it?** About 15 min per 100 kg for chopping, ~60 min for pellets, ~75 min for briquettes.
- **What if the residue is wet?** Above 20% moisture it's rejected. The station has a drying yard.
- **What if it breaks?** Spare blades, and the system alerts a technician every 200 working hours.
- **Is it safe?** Lid cut-off switch, hand guard, magnet for metal, emergency stop, dust extraction.

- **Is it built?** It's a detailed 3D design and a working software prototype. Building a pilot machine is the next step.

Business

- **How do you make money?** Margin on products, processing fees, and 10% of pickup fees.
- **How much does it cost?** About 200,000 THB per machine (150k–280k). The station depends on local land and budget.
- **When does it pay back?** Roughly 17 months at 400 kg/day and ~1.2 THB/kg margin. This is an illustration to be tested.
- **Why 1 baht per kg?** It beats burning (zero income) and leaves room for margin after processing.
- **Who pays for the pickup fund?** It comes from product margins, because pickup is part of our raw-material cost. It only applies when farmers sell to us.
- **Why 30%?** So the pickup never eats most of a small farmer's income.
- **What stops people gaming pickups?** Pickup needs at least 300 kg or a full load, and we suggest sharing trips with nearby farms.
- **Competitors?** Big biomass power plants buy residue but only near the plant and in bulk. We go to the village and to small farmers.

Impact & SDGs

- **Which SDGs?** 3, 8, 12, 13 (also 7 and 15).
- **How do you measure impact?** Every run is logged: kg not burned, PM2.5 avoided (~8 g/kg), farmer income, driver income. It's shown on the dashboard.
- **Is the 8 g/kg PM2.5 figure real?** It's an estimate; we'll replace it with a referenced research value.

Tech & law

- **Why LINE?** Farmers already use it every day. No new app to install.
- **What about personal data (PDPA)?** Separate consent page, only needed data, masked names in public, and farmers can withdraw consent and delete their data.
- **Can you sell the animal feed?** At first we use it in community farms. Selling it needs registration under Thai feed law.
- **Can the brand be protected?** Yes, "Rai Mai Phao" can be registered as a trademark. The code and 3D model are copyrighted automatically.

Hard ones

- **Biggest risk?** The numbers are untested. A pilot machine is our next step.
- **What would you do with funding?** Build one pilot machine in a Chiang Rai village, test yields and costs, then fix the numbers.
- **What did you learn?** The real problem isn't awareness, it's economics. Make the right choice the cheaper choice.

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9. Pre-opening checklist

- [ ] iPad on `/ipad/3d`, model loaded, English, full screen
- [ ] Monitor on `/present`, press **F**
- [ ] Phone: one test sale in LINE, receipt received
- [ ] Wi-Fi + hotspot backup · chargers · power strip
- [ ] Rehearse the 30-second and 4-minute versions with a timer

- [ ] Practise handing the iPad over and switching to **Business Model**

Links: sustainable-final.vercel.app · /ipad/3d · /ipad/bmc · /present